Patrick Thangarajah

Product Design | Prototyping | Testing & Validation | Simulation
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Summary

Recent MASc graduate in mechanical engineering with practical experience designing, building, and testing hardware systems, including sensor-integrated test setups and control circuitry. Brings simulation, DOE, and data analysis skills to support designs from concept through validation.

Skills

Technical Communication	Scientific literature reviews, technical reports, technical presentations, work instructions, SOPs
Design & Prototyping	SolidWorks, Onshape, 3D Printing, Lathe, Mill, Bandsaw, Drill Press
Controls & Instrumentation	Arduino, Sensors, Motors, Data Acquisition, Circuit Design, Altium, LTspice
Testing & Analysis	DOE, JMP, Minitab, MATLAB, Microsoft Excel, Troubleshooting

Education

Master of Applied Science in Mechanical and Mechatronics Engineering — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Research in occupant safety and biomechanics

Bachelor of Applied Science in Mechanical Engineering, Co-op Program — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2018 - May 2023*
- Biomechanics and Medical Devices Option

Experience

Graduate Research Student - Impact Mechanics and Material Characterization Group — *Waterloo, ON, Canada*
University of Waterloo — *Sep 2023 - Jul 2026*
- Communicated findings through a master's thesis, conference paper, technical reports, and six sponsor presentations
- Validated simulations against physical tests across two configurations and 10 response channels, achieving a 0.71 average ISO 18571 correlation score
- Ran a 104-configuration LS-DYNA DOE and used JMP to quantify geometry effects on occupant response, identifying a configuration with 16% lower peak chest deflection
- Automated seat belt fitting with LS-PrePost scripts, reducing repeated setup steps by 43% (7 to 4)

Teaching Assistant - MTE 120: Circuits — *Waterloo, ON, Canada*
University of Waterloo — *May 2024 - Aug 2024*
- Supported laboratory instruction for a 100+ student circuits course, troubleshooting hardware and providing feedback on lab work and exams

R&D Engineer (Part-time) — *Remote*
Echoscope — *May 2023 - Sep 2023*
- Authored five manufacturing and verification procedures covering PCB assembly, wiring, software setup, and functionality testing; all five were approved and used by the client
- Designed and iterated the bladder testing platform PCB, integrating pressure sensing, Arduino connections, power distribution, and motor and valve interfaces; refined component placement and routing and completed design-rule checks

Mechanical Engineering Intern — *Airdrie, AB, Canada*
GN Corporation — *Jan 2022 - Apr 2022*
- Developed standardized work instructions and SOPs by reviewing engineering drawings, customer specifications, and existing manufacturing processes
- Reduced fiberglass composite cutting waste by 50% through process analysis and proposed a material-handling workflow projected to reduce pick-and-place time by 50%

Engineering Intern — *Victoria, BC, Canada*
Breathe Global — *Jul 2021 - Aug 2021*
- Redesigned the analog control circuit for a negative-pressure breathing device in Altium, using LTspice simulation to select low-cost components
- Expanded the device's achievable breathing range from 70% to 88% of target

Projects

Soft Distensible Bladder Model - Capstone Project — *Vancouver, BC, Canada*
University of British Columbia — *Sep 2022 - Apr 2023*
- Designed and built the control and data-acquisition interface for a pressurized soft-bladder test system, enabling real-time pressure/volume monitoring and automated filling/voiding cycles
- Validated system performance using integrated pressure sensing and controls, demonstrating stable and repeatable operation up to 4.75 kPa internal pressure

Conference Presentations and Publications

- **Thangarajah, P.** (2026). *Assessment of seat belt anchor position on rear seat occupant kinematics in frontal impact using a validated THOR-5F finite element model*. Master of Applied Science thesis, University of Waterloo, Waterloo, ON, Canada. hdl.handle.net/10012/23684
- **Thangarajah, P.**, Watson, B., Tang, K., Tylko, S., & Cronin, D. (2025). *Analysis of rear seat shoulder belt anchor position on kinematics of small-stature female occupants in frontal impact*. Injury Biomechanics Symposium, The Ohio State University, Columbus, OH, USA, May 18–21, 2025.